

Foundation Engineering

1. Foundations can be broadly classified under _____

a) Shallow foundation and Deep foundation b) Pile foundation

c) None of the mentioned d) All of the mentioned View Answer

Answer: a

Explanation: Foundations may be broadly classified under two heads: shallow foundations and deep foundations.

2. A foundation is said to be shallow if its depth is _____ than its width. a) Equal to and Less than

b) Greater than

c) None of the mentioned d) All of the mentioned View Answer

Answer: a

Explanation: According to Terzaghi, a foundation is shallow if its depth is equal to or less than its width. In the case of deep foundations, the depth is equal to or greater than the width.

3. Which of the following is a type of shallow footing? a) Spread footing

b) Pile foundation c) Pier foundation

d) Well foundation View Answer

Answer: a

Explanation: The shallow foundations are of the following types: spread footing (or simple footing), strap footing, combined footing, and mat footing or raft footing.

4. Which of the below is the most commonly used shallow foundation? a) Strap footing

b) Spread footing

c) Combined footing d) Raft footing View Answer

Answer: b

Explanation: In spread footing, load is transmitted through an isolated column or wall to the subsoil; hence this is most common type of foundation.

5. The pressure intensity beneath the footing depends upon _____

a) Rigidity of the footing b) Soil type

c) Condition of soil

d) All of the mentioned View Answer

Answer: d

Explanation: Both from observations as well as the analytical studies from elasticity, it is known that the pressure distribution beneath footing is not uniform and it depends on the rigidity of footing, the soil type, and the condition of the soil.

6. Once the pressure distribution is known _____ in the reinforced concrete footing can be calculated. a) Bending moment and Shear force

b) Bearing pressure

c) None of the mentioned

d) All of the mentioned View Answer

Answer: a

Explanation: Knowing the pressure distribution is known, the bending moment and shear force can be calculated, and the thickness of the structural member of the member can be calculated using properties of reinforced concrete.

7. In conventional design, allowable bearing capacity should be taken smaller than which of the following value? a) Safe bearing capacity and Allowable bearing pressure

b) The pressure intensities beneath the footing c) None of the mentioned

d) All of the mentioned View Answer

Answer: a

Explanation: In conventional design, the allowable bearing capacity should be taken as the smaller of the following two values: i) the safe bearing capacity based on ultimate capacity, and ii) the allowable bearing pressure on tolerable settlement.

8. In cohesive soil, the pressure distribution beneath the footing is _____

a) Linear

b) Non linear c) Zero

d) None of the mentioned View Answer

Answer: a

Explanation: When a footing rests on cohesive soil, the edge stresses may be very large, but the pressure distribution may be considered to be linear.

9. When do strap footings are used in foundation? a) To transfer load of an isolated column

b) Distance between the columns are long c) Two column loads are unequal

d) All of the mentioned View Answer

Answer: b

Explanation: A strap footing may be used where the distance between the columns is so great that the combined trapezoidal footing becomes quite narrow, with a high bending moment.

10. When two column loads are unequal, which of the possible footing can be provided? a) Strap footing

b) Raft footing

c) Trapezoidal combined footing d) Mat footing

View Answer

Answer: c

Explanation: When the two columns load are unequal, with the outer column carrying a heavier load, and when there is space limitation beyond the outer column, a trapezoidal footing is provided

n SPT, The test uses a thick-walled sample tube, with an outside diameter

A 45 mm B 50 mm C 55 mm D 53 mm

Correct Choice ☐ B

2) In SPT, we terminate the test for no. of blows which are obtained to drive the required 12-in

A 70 blows B 80 blows C 90 blows

D 100 blows

Correct Choice ☐ D

3) Gravels are Lines more previous than clays

A 10,000 B 50,000 C 1 million

D 30,000

Correct Choice ☐ C

4) The design of dams vary much based upon the permeability of soils used:

A Concrete dam B Earth Dam C Coffee Dam D Steel Dam

Correct Choice ☐ B

5) Units for K (co-efficient of permeability) are :

Am^3/sec

B $[\text{cm}]^2/\text{sec}$

$\text{Cm}/[\text{sec}]^2$

Dcm/sec

Correct Choice ☐ D

6) Major principle stress is given by $\sigma_1 = 40 \text{ KN/m}^2$ and Minor

principle stress is given by $\sigma_3 = 20 \text{ KN/m}^2$, shear strength is given by :

$$Z = (\sigma_1 - \sigma_3)/2$$

A 20 KN/m^2 B 40 KN/m^2 C 60 KN/m^2 D 10 KN/m^2

Correct Choice ☐ D

7) Unconfined compressive strength is given by 120 KN/m^2 , what will be the unconfined shear strength or cohesion?

$$C = q_u/2$$

A 60 KN/m^2 B 80 KN/m^2 C 99 KN/m^2 D 120 KN/m^2

Correct Choice ☐ A

8) According to ASTM D2573, in Field Vane Shear test with the vane in position, apply the torque to the vane at a rate which should not exceed _____.

A 1°/sec B 4°/sec C 1°/min D 4°/min

Correct Choice ☐ A

9) According to ASTM D2573, Field Vane Shear test in case of very soft clays the time to failure may be as much as _____.

A 3 – 5 min

B 5 – 8 min C 10 – 15 min D 15 – 18 min

Correct Choice ☐ C

10) A structure is erected on impervious clay whose thickness is 12m. Drainage is possible both at upper and lower surfaces. Coefficient of consolidation is 0.015 cm² per minute. For attaining 50% consolidation with a time factor of 0.20, the number of days required??

A 3123 B 3333 C 3233

D 3313

Correct Choice ☐ B

11) A Standard proctor compaction test performed on sample of crushed limestone ($G_s = 2.70$) obtained a maximum dry unit weight of 90 Pcf at OMC. A field compacted sample showed a moisture of 28% and unit weight of 103.7 pcf. Find the relative compaction (RC) = ?

A 70% B 80% C 90% D 100%

Correct Choice ☐ C

12) The moist unit weight of a soil is 16.5 kN/m³. Given that the $w = 15\%$ and $G_s = 2.70$, Find the porosity $n = ?$

A 46% B 48% C 50% D 52%

Correct Choice ☐ A

13) Number of blows $N=32$, Find allowable bearing capacity in KPa =? using B

A 14 B 12 C 10 D 8

Correct Choice ☐ D

14) The permissible settlement of a shallow foundation on a rock evaluation of safe bearing pressure from a plate load test is taken as

.....

A 12mm B 25mm C 40mm D 50mm

Correct Choice ☐ D

15) In loose coarse gravel, the split spoon (SPT) tends to slide into the

....., and resistance is observed.

A large voids, low penetration B small voids, large penetration C small voids, small penetration D large voids, large penetration

Correct Choice ☐ A

16) Expression for “Q allowable” using Meyerhoff’s equation in FPS system for $B < 4\text{ft}$ is

$\frac{AN}{4} K_d$ B $\frac{N}{2.5} K_d$

CN/8 Kd D 12NKd

Correct Choice ☐ A

17) The contact pressure of flexible footing on non-cohesive soils is:

A Uniform throughout.

B More in the centre than at edges. C Less at the centre than at edges. D Not on the footing.

Correct Choice ☐ B

18) In case of lateral loads or moments, the foundation should also be checked to be safe against sliding and overturning. The FOS shall not be less than

A 1.75 against sliding

B 2.0 against overturning. C 3.0 for wind seismic D a, b both

Correct Choice ☐ D

19) For Strip Spread Footings

- Length (L) to width (B) ratio,

A $L/B < 10$ B $L/B \geq 10$ C $L=B$

D B is very small

Correct Choice ☐ B

20) In shallow foundation, the contribution of side shear stress is

A Maximum B Minimum C Negligible

D Average

Correct Choice ☐ C

21) Which of the following statement is correct?

A The settlement of rigid footing on cohesion less soil is uniform throughout.

B The settlement of flexible footing on cohesive soil is less in the center than at the edges.

C The settlement of flexible footing on cohesion less soil is more in the center than at the edges.

D The settlement of rigid footing on cohesive soil is uniform throughout.

Correct Choice ☐ A

22) In shallow foundation if $R_w = 1$ & $R_w' = 0.5$ than where the water table lies:

A At base of footing B Below the footing C At the ground level D Any where at the mid

Correct Choice ☐ A

23) If $Q_u = Q_s + Q_b$ where Q_s = Shaft resistance, Q_b = Bearing resistance

And Q_s is greater than Q_b the pile is classified as

a.skin friction pile

b.End bearing pile

c.Cast in –situ pile

d.Driven pile

Correct Choice ☐ A

24)Which type of pile is used near sea to protect harbor just by absorbing the effect of floating objects

a.Anchor piles

b.Fender piles

c.Batter piles

d.Sheet piles

Correct Choice ☐ B

25)For over consolidated soil the value of OCR is

a.equal to one

b.greater than one

c.can't be measured

d.smaller than one

Correct Choice ☐ B

26)Find the approximately No. of Standard Penetration's (N) by

using Cone Penetration Resistance (q_c) of 20 MPa, if 'k' factor is 0.3 and the soil is silty sand; ($q_c=kN$)

A66 B 77 C 88

D99

Correct Choice ☐ A

27)Max. Overburden pressure (q') at the tip of the pile can be calculated by;

A $2\gamma D_f$

B $(1/2)\gamma D_f$ C γD_f

D $(3/2) \gamma D_f$

Correct Choice ☐ C

28)If the ultimate load carrying capacity(Q_u) of certain pile is 500

kips, the shaft resistance (Q_s) is 415kips and bearing resistance (Q_b) is 40 kips then comment about the nature of the pile;

A Cast in situ pile B Driven Pile

C Sheet pile

D None

Correct Choice ☐ B

29) If the shaft resistance $Q_s = 300,000$ lbs and bearing resistance Q_b is equal to 30,000 lbs then ultimate load carrying capacity Q_u is equal to;

A 300 kips B 330 kips C 310 kips

D None

Correct Choice ☐ B

30) For clayey soil " F_s " will be equal to "where F_s is pile unit surface skin friction"

A $\alpha c + q_k \tan \delta$

B $q_k \tan \delta$

C αc

D none of above

Correct Choice ☐ C

31) If a soil is fully saturated having void ratio 0.67 what will be the porosity?

A 40% B 42% C 44% D 46%

Correct Choice ☐ A

32) D₆₀ is 9.6mm, D₃₀ is 2mm and D₁₀ is 0.16mm what is value of C_c (coefficient of curvature)?

a) 2.4

b) 2.5

c) 2.6

d) 2.7

Correct Choice ☐ C

33) For construction of an embankment, if max dry unit wt is 120.5lb/ft³, optimum moisture content 13%, dry density 118lb/ft³, actual field moisture content 12.9%, the relative compaction will be?

A-94%

B-96%

C-98%

D-100%

Correct Choice ☐ C

34) The total weight to be arranged for a pile load test in compression

should be:

A equal to the anticipated maximum test load

B at least 100% greater than the anticipated maximum test load
C at least 50% greater than the anticipated maximum test load
D at least 10% greater than the anticipated maximum test load

Correct Choice ☐ D

35) A plate (0.3 x 0.3 m plate size) load test performed on clayey soil gives ultimate settlement of 2.2 mm. The ultimate settlement of 2 x 2 m isolated footing will be:

A 18.65 mm B 12.50 mm C 14.67 mm D 16.39 mm

Correct Choice ☐ C

36) Usually for driven piles, the value of lateral earth pressure co-efficient "K" is

A ≥ 1 B ≤ 1 C 1.75

D zero

Correct Choice ☐ A

37) For very hard-clay, value of cohesion reduction factor " α " is

A 1

B 0.3 C 0.4 D 3

Correct Choice ☐ C

38) In vane shear test if ($0.6 < I_d < 1.8$) then the soil type will be:

A Sand

B Silt C Clay

D Gravely soil

Correct Choice ☐ B

39) In vane shear test initially ΔA , ΔB (before inserting the blade) must be in the Ranges:

A $\Delta A = 15$ to 30 kPa B $\Delta A = 5$ to 30 kips C $\Delta A = 5$ to 30 kPa D $\Delta A = 20$ to 30 kPa

Correct Choice ☐ C

$\Delta B = 15$ to 80 kPa $\Delta B = 5$ to 80 kips $\Delta B = 5$ to 80 kPa $\Delta B = 20$ to 80 kPa

40) Foundation on loess in situ dry densities range from?

A 10 to 14.6 kN/m³ B 10 to 15.5 kN/m³ C 10 to 16.5 kN/m³ D 10 to 17.5 kN/m³

Correct Choice ☐ C

41) When footings are to be placed adjacent to an existing structure, the line from the base of the new footing to the bottom edge of the existing footing should be

A 30 degree B 45 degree C 60 degree D 90 degree

Correct Choice ☐ B

42) The vane head should be turned five complete rotations at a speed of approximately _____ seconds per rotation, and steps 3.6 to 3.8 repeated

A10 B 30 C 60

D5

Correct Choice ☐ A

43)The backfill used in Reinforced earth structures should contain less than _____ fines content as required by the AASHTO Specifications for Highway Bridges.

A15 % B 18% C 25% D 28%

Correct Choice ☐ A

44)In CPT test, the rate at which we push the instrumented con tip into the ground is

A2 centimeters/second B 4 centimeters/second C 5 centimeters/second D 3 centimeters/second

Correct Choice ☐ A

45)In Case of SPT , the graph between Depth of influence and Breadth has behavior

AInversly proportional B Constant line

CDirectly proportional D Non uniform behaviour

Correct Choice ☐ C

46)Geotechnical investigation of soil resulted in SPT value of $N=30$. If square footing dimensions are confined to be 5 feet. Suggest allowable bearing capacity for permissible settlement of 1 inch using Teng Method.

A6 K/ft² B 7 K/ft² C 9 K/ft² D 8 K/ft²

Correct Choice ☐ B

47) Geotechnical investigation of soil resulted in spt value $N=25$, allowable bearing capacity of soil was found to be 6 k/ft^2 . Suggest suitable dimensions of square footing using Meyerhof Method for ($B > 4\text{ feet}$)?

A 5 feet B 6 feet C 7 feet D 8 feet

Correct Choice ☐ A

48) Geotechnical investigation of soil resulted in SPT value of $N=30$. If square footing dimensions are confined to be 5 feet. Suggest allowable bearing capacity for permissible settlement of 1 inch using Teng Method.

A 6 K/ft^2 B 7 K/ft^2 C 9 K/ft^2

D 8 K/ft^2

Correct Choice ☐ B

49) The un-drained shear strength of cohesive soil and SPT N is controlled by

A Plasticity B Sensitivity C Fissuring

D All of Above

Correct Choice ☐ D 50) Black cotton soils are:

A Residual Soils B Organic Soils C Expansive Soils D Ash Soils

Correct Choice ☐ B

51)The expansion of soil due to shear at a constant value of pressure is called.....

A Apparent cohesion B Dilatancy

C true cohesion

D Consistency

Correct Choice ☐ B

52)Inorganic clays of high plasticity and having high inherent swelling capacity have liquid limit exceeding..... % and plastic index

over

A 20% , 10

B 40% , 30 C 30% , 20

D 50% , 30

Correct Choice ☐ D

53)The best way to avoid damage from expansive soils is :

A To give a shallow foundation B To give a pile foundation

C To give foundation beneath the water zone D To use of cobbles

Correct Choice ☐ C

54) Which of the following is deep foundation

A Mat foundation B Strip Footing C Rock Anchor

D Caisson

Correct Choice ☐ D

1. How many types of foundations are there based on depth? a) 3

b) 4 c) 5 d) 2

View Answer

Answer: d

Explanation: The 2 types are shallow and deep foundations. Shallow foundation transfers the load to very near earth. Deep foundations are used for high rise buildings to bear heavy loads.

2. The figure below represents:

a) Isolated footing

b) Wall footing

c) Strap footing

d) Mat foundation View Answer

Answer: b

Explanation: Wall footing or strip footing is a continuous strip of concrete that serves to spread weight of load bearing wall.

3. _____ footing is used in load bearing masonry construction. a) Isolated

b) Strap c) Strip d) Pile View Answer

Answer: c

Explanation: The strip footing bears a whole load of an entire wall. It is a continuous footing that runs below the wall.

4. How many types of combined footing are possible? a) 2

b) 3

c) 4

d) 5

View Answer

Answer: a

Explanation: Combined footing is provided when 2 columns are very close by. They can be rectangular or trapezoidal in section.

5. CPRF stands for:

a) Combined Plain Round Foundation

b)Connected Pile Round Foundation

c)Combined Pile Raft Foundation

d)Corrosion Proof Raft Foundation View Answer

Answer: c

Explanation: CPRF uses the combination of pile and raft foundation as the name suggests. It is used as a foundation for high rise buildings. It is more powerful, strong, load bearing, safe than raft or pile when used.

6.Micropiles find its main application in retaining walls. a) True

b) False View Answer

Answer: b

Explanation: Micropiles are used for under pinning. They are used in highways, bridges and transmission tower projects. Sheet piles are extensively used in retaining walls.

7.Steining is a component of which of the below type of foundation? a) Pile

b) Strap c) Isolated

d)Well View Answer

Answer: d

Explanation: Well foundation is a type of deep foundation. It is shaped like a well. The well is dug, filled with sand/concrete. Steining is the wall provided to the well and it is built over a wedge shaped portion, called well curb.

8. Pier foundation is also called: a) Caisson

b) Box c) Bridge d) Girder View Answer

Answer: a

Explanation: Caisson foundation is often used in the construction of bridge piers, hence it is also called pier foundation. Caisson can be floated to the site and sunk to the required position. It is used under water also.

9. The figure below represents _____ piles.

a) Load bearing

b) End bearing

c) Friction

d) Sheet

View Answer

Answer: c

Explanation: Friction piles are usually used in construction to provide underground support to structures. They work on static friction developed between soil surface and the pile.

10. Which of the below is not a preliminary consideration for building a foundation? a) Bearing capacity of soil

b) Ground water condition c) Settlement control

d) Soil organisms View Answer

Answer: d

Explanation: Bearing capacity of soil, ground water conditions and settlement control are all very important parameters to be considered while selecting the right foundation. Every soil contains organisms, foundation should be able to withstand their activities

1. Earth embankments or slopes are commonly required for which of the following purpose? a) Railways

b) Earth dams c) Road ways

d) All of the mentioned View Answer

Answer: d

Explanation: Earth embankments are commonly required for railways, roadways, earth dams, levees and river training works.

2. The failure of slopes may take place due to _____

a) Forces between the soil particle and High water content b) Action of gravitational force

c) None of the mentioned d) All of the mentioned View Answer

Answer: a

Explanation: The failure of slopes takes place mainly due to the action of a gravitational force which tends to pull the upper portions of the soil mass downwards to a nearly more level surface.

3. Slopes are classified into _____ types. a) 2

b) 3 c) 4 d) 5

[View Answer](#)

Answer: a

Explanation: Slopes may be of two types: i) infinite slopes and ii) finite slopes.

4. Which of the following is an example of slopes extending to infinity? a) Inclined face of Earth dams

b) Embankments c) Cuts

d) None of the mentioned [View Answer](#)

Answer: d

Explanation: Slopes extending to infinity do not exist in nature.

5. Analysis of stability of slopes is used for determining _____

a) Shearing strength and Stressed internal surface b) Properties of the soil

c) None of the mentioned d) All of the mentioned [View Answer](#)

Answer: a

Explanation: An analysis of stability of slopes consist of determining the most severely stressed internal surface and the magnitude of the shearing stress to which it is subjected and determining the shearing stress along the surface.

6. If the soil properties for all identical depths below the surface are constant, it is a _____

a) Finite slope b) Infinite slope

c) Planar failure surface d) All of the mentioned View Answer

Answer: b

Explanation: If a slope represents the boundary surface of a semi-infinite soil mass, and the soil properties for all identical depths below the surface are constant, it is called an infinite slope.

7. A long natural slope of cohesion-less soil is inclined at 12° to the horizontal. What will be the factor of safety of the slope if $\phi = 30^\circ$?

a) 1.6 b) 2.7 c) 0.13 d) 0.4 View Answer

Answer: b

Explanation: Given $\phi = 30^\circ$ and $i = 12^\circ$

Formula, $F = \tan \phi / \tan i$

$$F = \tan 30^\circ / \tan 12^\circ$$

$$F = 2.72.$$

8. Factor of safety with respect to height is given by _____ equation. a) $FC = HC / H$

b) $FC = C / C_m$

c) $FC = \tau_f / \tau$

d) $FC = \tau_c / \tau$ View Answer

Answer: a

Explanation: Since the factor of safety FC with respect to cohesion, also represent the factor of safety with respect to height $FC = C / C_m = HC / H$

Where, C_m is the mobilized cohesion at a depth H .

9. In the limiting case of stability, the angle of slope is referred to as _____

a) Angle of deviation b) Angle of repose

c) Angle of unstable slope d) All of the mentioned View Answer

Answer: b

Explanation: In the limiting case of stability, the angle of slope is known as the angle of repose ϕ .

10. The factor of safety against sliding is given by _____ equation. a) $F = HC / H$

b) $F = \tau_f / \tau$ and $F = \tan \phi / \tan i$ c) None of the mentioned

d) All of the mentioned View Answer

Answer: b

Explanation: The factor of safety against sliding is given by equation:

$$F = \tau_f / \tau = \tan \phi / \tan i$$

1. In hydrometer method, the sampling depth h is kept at a constant of _____

a) 14 cm b) 9 cm c) 8 cm d) 10 cm View Answer

Answer: d

Explanation: The sampling depth (h) in hydrometer method is taken as 10 cm.

2. In calibration of hydrometer reading, the reduced reading are designated as _____

a) RS b) RH c) RR d) RL View Answer

Answer: b

Explanation: The reducing reading is designated as RH, which is used for calibration of hydrometer.

3. The dispersing agent correction in hydrometer reading is always _____

a) Positive

b) Both negative and positive c) Negative

d) Equal View Answer

Answer: c

Explanation: The addition of dispersing agent in water increases its water level, therefore the dispersing correction is always negative.

4. The hydrometer method differs from pipette method on the basis of which of the following? a) Principle of test

b) Taking the observation

c) Method of procedure

d) None of the mentioned View Answer

Answer: b

Explanation: In sedimentation analysis the hydrometer method differs from the pipette method in the method of taking the observation.

5. The corrections which are applied to the hydrometer reading, in hydrometer method is _____

a) Meniscus correction and Dispersing agent correction b) Magnitude correction

c) All of the mentioned d) None of the mentioned View Answer

Answer: a

Explanation: Both meniscus and dispersing agent correction are applied to a hydrometer reading.

6. The meniscus correction in hydrometer reading is _____

a) Always positive b) Always negative c) Negative sometime d) Positive sometime View Answer

Answer: a

Explanation: The actual reading to be taken at water level will be more; hence the meniscus correction will be always positive.

7. Which of the following is the correct hydrometer reading equation? a) $R = R_h' + C_m \pm C_L - C_d$

b) $R = R_h' + c_m + C_L - C_d$ c) $R = C_m \pm R_m - C_d + C_L$

d) $R = R_h + C_m \pm C_d + C_L$ View Answer

Answer: a

Explanation: According to the derived equation of hydrometer the equation is

$$R = R_h' + C_m \pm C_L - C_d.$$

8. The corrected hydrometer reading for meniscus is _____

a) $R = R_h' + C_m$ b) $R = R_h' + C$

c) $R = R_h' + C_m \pm C_L - C_d$ d) None of the mentioned View Answer

Answer: b

Explanation: The three corrections C_m , C_L , C_d can be combined into one correction C_m and the equation can be written as $R = R_h' + C_m$.

9. The visibility of the soil suspension in hydrometer method is _____

a) Transparent b) Translucent c) Opaque

d) None of the mentioned View Answer

Answer: c

Explanation: Since the density of the soil suspension is high, the visibility will be opaque.

10. The volume of suspension taken in the hydrometer test is _____

a) 500 b) 120 c) 100

d) 1000 View Answer

Answer: d

Explanation: The quantity of dry soil and dispersing agent is taken as double of pipette which is 500 ml

1. When the water table is above the base of the footing, the submerged weight ' γ ' can be used to compute _____

a) Effective pressure and Surcharge b) Pore pressure

c) None of the mentioned d) All of the mentioned View Answer

Answer: a

Explanation: When the water table is above the base of the footing, the submerged weight ' γ ' should be used for the soil below the water table for computing the effective pressure or the surcharge.

2. Which of the following is an essential difference between Vesic's and Hansen's procedure? a) Different values of N_y and Variation on some of Hansen inclination

b) Different value of N_q and N_c c) None of the mentioned

d) All of the mentioned View Answer

Answer: a

Explanation: The essential difference in Vesic's and Hansen's procedure are i) use of slightly different values of N_y and ii) a variation on some of Hansen's inclination, base and ground factors.

3. The equation $N_c = (N_q - 1) \cot \phi$, have been adopted by _____

a) Terzaghi and Peck b) Hansen

c) Vesic

d) All of the mentioned View Answer

Answer: d

Explanation: The two equations $N_c = (N_q - 1) \cot \phi$ and $N_q = \tan^2 (45 + \phi/2) \times \tan \phi$ have been adopted by i) Terzaghi and Peck ii) Hansen iii) Vesic iv) Meyerhof and Bureau of Indian standards.

4. Which of the following N factors has the widest range of values? a) N_c

b) N_q c) N_γ

d) All of the mentioned View Answer

Answer: c

Explanation: The N_γ factor has the widest suggested range of any value of N factors. A literature value reveals that for $\phi = 40^\circ$, $38 \leq N_\gamma \leq 192$.

5. Which of the following term does not contribute to q ? a) N_c

b) N_γ c) N_q

d) All of the mentioned View Answer

Answer: b

Explanation: The N_γ term does not contribute significantly to q_f so that almost any reasonable value can be used. Bureau of Indian standards (BIS) recommends using the value of N_γ , given by Vesic.

6. Meyerhof's extended the analysis of plastic equilibrium of a surface footing to _____

a) Shallow foundation and Deep foundation b) Inclined foundation

c) None of the mentioned

d) All of the mentioned View Answer

Answer: a

Explanation: M Meyerhof's extended the analysis of plastic equilibrium of a surface footing to Shallow and Deep foundation according to both Terzaghi and Meyerhof analysis.

7. In the Bowles' method for finding the effect of water table, which of the following factor is used for soil in the wedge zone? a) Water reduction factor

b) Effective unit weight c) Overburden pressure d) Average unit weight View Answer

Answer: b

Explanation: In the third method (Bowles, 1988), no water reduction factor is used, but effective unit weight (γ_e) is used for the soil in the wedge zone.

8. Which of the following equation has been recommended by Indian standard for finding reduction factor in water table? a) $R_w = 0.5 (1 + z^2/B)$

b) $R_w = (1 + z^2/B)$

c) $R_w = c N_c + \bar{\sigma}_v N_q + 0.5 \gamma B N_\gamma$ d) $R_w = z^2 / B$

View Answer

Answer: a

Explanation: $R_w = R_{w2} = 5 \left(1 + \frac{z}{B} \right)$ has been recommended by Indian standard at a depth D_1 below the ground level.

9. The value of N_c and N_q are same for, which of the following methods? a) Meyerhof

b) Hansen

c) Vesic

d) All of the mentioned View Answer

Answer: d

Explanation: The value of N_c and N_q are same for Meyerhof, Hansen and Vesic method, while subscripts for N_γ identify the author, i.e. $N_\gamma(H)$ = valued by Hansen etc.

10. The factor $N_\gamma(M)$ identifies which of the following author? a) Vesic

b) Meyerhof c) Hansen d) Terzaghi View Answer

Answer: b

Explanation: The factor $N_\gamma(M)$ gives values by Meyerhof, as it has the subscript (M) in it.

1. An analysis of the condition of complete bearing capacity failure is usually termed as _____

a) General shear failure b) Terzaghi's analysis c) Bearing failure

d) All of the mentioned View Answer

2.The concept of analysis of bearing capacity failure was first developed by _____

a)Terzaghi

b)Meyerhof

c)Prandtl

d)Darcy View Answer

Answer: c

Explanation: The concept of failure analysis was first developed by Prandtl, and later extended by Terzaghi, Meyerhof and others.

3.For purely cohesive soil, the bearing capacity is given by which of the following equation? a) $q_f = 5.7 c + \bar{\sigma}$

b) $q_f = c + \bar{\sigma}$ c) $q_f = 5.7 c$

d) All of the mentioned View Answer

Answer: a

Explanation: For purely cohesive soil the bearing capacity is $q_f = c N_c + \bar{\sigma} N_q = 5.7 c + \bar{\sigma}$

Where $\bar{\sigma} = \gamma D$ if the water table is below the base of the footing.

4.The parameters N_c , N_q , N_γ in the equations of bearing capacity failure are known as _____

a) Constant head

b) Bearing capacity factors c) Effective pressure

d) Load intensity View Answer

Answer: b

Explanation: The parameters N_c , N_q , N_γ are the dimensionless numbers, known as bearing capacity factors depending only on the angle of shearing resistance of the soil.

5. For purely cohesive soil, local shear failure may be assumed to occur when the soil is _____

a) Medium to soft b) Soft to medium c) Hard

d) All of the mentioned View Answer

Answer: b

Explanation: For purely cohesive soil, local shear failure may be assumed to occur when the soil is soft to medium, with an unconfined compressive strength $q_u \leq 100 \text{ kN/m}^2$.

6. Which of the following is a limitation, of assumption in Terzaghi's analysis? a) ϕ changes when the soil is compressed and strip footing has a rough base b) Soil is homogeneous

c) None of the mentioned d) All of the mentioned View Answer

Answer: a

Explanation: As the soil compress, ϕ changes; slight downward movement of footing may not develop fully the plastic planes.

7. Which of the following are original Terzaghi values for N_γ ? a) 34° and 48°

b) 60°

c) None of the mentioned d) All of the mentioned View Answer

Answer: a

Explanation: The values of N_γ for ϕ of 34° and 48° are the original Terzaghi values which were used by Bowles to back compute K_γ .

8. According to the assumptions in Terzaghi's analysis, the soil is _____

a) Homogeneous and Isotropic b) Non Homogeneous

c) None of the mentioned

d) All of the mentioned View Answer

Answer: a

Explanation: In Terzaghi's analysis the soil is homogeneous and isotropic and its shear strength is represented by Coulomb's equation.

9. The Terzaghi's general bearing capacity equation is represented as _____

a) $q_f = 5.7 c + \bar{\sigma}$

b) $q_f = c N_c + \bar{\sigma} \cdot N_q + 0.5 \gamma B N_\gamma$ c) $q_f = c N_c + \bar{\sigma} \cdot N_q$

d) $q_f = c N_c$ View Answer

Answer: b

Explanation: " $q_f = c N_c + \bar{\sigma} N_q + 0.5 \gamma B N_\gamma$ " is known as Terzaghi's general bearing capacity equation for a continuous footing.

10. Local shear failure generally occurs in _____

a) Dense sand

b) Non-cohesive soil c) Loose sand

d) All of the mentioned View Answer

Answer: c

Explanation: Local shear failure generally occurs in loose sand while general shear failure occurs in dense sand. advertisement

11. The curve for N_q and N_γ for the transition state from $\phi = 28^\circ$ and $\phi = 38^\circ$ was given by _____

a) Peck b) Hanson

c) Thorn burn

d) All of the mentioned View Answer

Answer: d

Explanation: Peck, Hansen, and Thorn burn gave curves for N_q and N_γ for the transition state from $\phi = 28^\circ$ and $\phi = 38^\circ$, in which they assumed general shear failure when $\phi > 28^\circ$.

12. Terzaghi's bearing capacity equation is not applicable for _____

- a) Depth effect and Inclination factor
 - b) Narrow slope
 - c) None of the mentioned
 - d) All of the mentioned
- View Answer

Answer: a

Explanation: Terzaghi assumed the value of angle $\psi = \phi$, which is not true. Since footings are normally rough, ψ has been found close to $45^\circ + \phi/2$ than to ϕ , thus Terzaghi's bearing capacity equations do not have provision for including depth effects, inclination factors, etc

1. IS code recommends a bearing capacity equation, which is similar to _____

- a) Vesic
 - b) Meyerhof and Brinch Hansen
 - c) None of the mentioned
 - d) All of the mentioned
- View Answer

Answer: b

Explanation: IS code (IS : 6403-1981) recommends a bearing capacity equation which is similar in nature to those given by Meyerhof and Brinch Hansen.

2. According to IS code, there are _____ types of failures of soil support beneath the foundation. a) 2

b) 4

c) 3

d)5

[View Answer](#)

Answer: c

Explanation: The code recognizes, depending upon the deformation associated with the load and the extent of development of failure surface, three types of failures of soil support beneath the foundations which are general, local and punching shear failure.

3.The bearing capacity equation for strip footing as given IS standard, can be modified on the basis of _____

a) Shape of the footing b) Type of soil

c) Bearing capacity

d) All of the mentioned [View Answer](#)

Answer: a

Explanation: The bearing equations applicable for strip footing shall be modified to take into account, the shape of the footings, and inclination of loading, depth of embankment and effect of water table.

4.The value of shape factor S_c , S_q and S_γ for circular base is _____

a) 1.3, 1.2, 0.8 b) 1.3, 1.2, 0.6 c) 1.0, 1.0, 1.0 d) 1.2, 1.0 and 1.3 [View Answer](#)

Answer: b

Explanation: According to IS (6403 -1981) standards, the shape factor for circle base is 1.3, 1.2 and 0.6 for S_c , S_q and S_γ respectively.

5.The depth factor can be applied to footing only when _____

- a) Back filling is compacted
- b) Shape factors are not used
- c) The base of the footing is circular
- d) All of the mentioned

View Answer

Answer: a

Explanation: The depth factor can be applied only the back filling is done with proper compaction.

6.The effect of the water table is taken into account for bearing capacity in the form of _____

- a) Depth factor
 - b) Inclination factor
 - c) Correction factor
 - d) Shape factor
- View Answer

Answer: c

Explanation: The effect of the water table is taken into account in the form of correction factor W' applied to the wedge term.

7.If the water table is located at the depth D , then the value of W' is taken as _____

- a) 1
- b) 0.5
- c) 2
- d) 0

[View Answer](#)

Answer: b

Explanation: If the water table is located at the depth D or likely to rise at the base of the footing or above, then the value of W' shall be taken as 0.5.

8. For cohesion less soils, having $c=0$ Indian standard code recommends that the bearing capacity can be calculated based on

a) Relative density

b) Standard penetration resistance value

c) Static cone penetration

d) All of the mentioned

[View Answer](#)

Answer: d

Explanation: For cohesionless soils ($c=0$), Indian standard code recommends that the bearing capacity can be calculated i) based on relative density or ii) based on standard penetration resistance value, and iii) based on static cone penetration test.

9. The standard penetration resistance is determined at a number of selected points at intervals of

a) 80 cm b) 75 cm c) 100 cm d) 10 cm [View Answer](#)

Answer: b

Explanation: Indian standard recommends that the standard penetration resistance is determined at a number of selected points at intervals of 75 cm in the vertical direction or change of strata if it takes place earlier.

10. The ultimate bearing capacity for cohesive soil can be calculated using the equation _____

a) $q_{nf} = c N_c$

b) $q_{nf} = c N_c$ c) $q_{nf} = N_c$

d) $q_{nf} = N_c$ View Answer

Answer: b

Explanation: The net ultimate bearing capacity after construction on fairly saturated homogeneous cohesive soils can be calculated from the expression:

$q_{nf} = c N_c$ where, $N_c = 5.14$.

advertisement

11. The value of c in the equation, $q_{nf} = c N_c$ used for finding bearing capacity of cohesive soil can be calculated by

a) Unconfined compression strength test b) Static cone penetration test

c) Relative density test d) All of the mentioned View Answer

Answer: a

Explanation: The value of cohesion c can be obtained from unconfined compression strength test. Alternatively, cohesion c may be determined by static cone resistance q_c using the empirical relation between q_c and c

1. _____ is used to indicate the maximum load per unit area which the soil will resist safely without displacement. a) Bearing power

b) Maximum safe bearing capacity c) Strength of soil

d) Allowable bearing pressure View Answer

Answer: a

Explanation: The Load of the structure is ultimately coming on the soil and hence it is of utmost importance to know the strength and behaviour of the soil. Dividing the ultimate bearing power of soil by a factor of safety, the safe bearing capacity of soil is obtained.

2. The term _____ on soil is used to indicate the maximum pressure which can be exerted on soil while taking into account shear failure, settlement and the ability of structure to resist settlement.

a) Bearing capacity

b) Net pressure intensity

c) Allowable bearing pressure

d) Maximum safe bearing capacity View Answer

Answer: c

Explanation: The value of Allowable Bearing Pressure on soil depends on the nature of ground and the type of building. The maximum safe bearing capacity of soil is generally more than the allowable bearing pressure on soil.

3.The maximum differential settlement should not exceed _____ mm in case of foundation on sandy soil. a) 100

b) 40 c) 50 d) 25

[View Answer](#)

Answer: d

Explanation: The total amount of settlement should be generally limited between 40 mm and 100 mm. The term differential settlement indicates the relative settlement of the adjacent portion of the structure.

4.The maximum differential settlement should not exceed _____ mm in case of foundation on clayey soil. a) 40

b) 25 c) 100 d) 50

[View Answer](#)

Answer: a

Explanation: For ordinary frame structures of concrete, the permissible angular distortion is $1/500$ i.e. one horizontal to 500 vertical and the desirable value is $1/1000$ i.e. one horizontal to 1000 vertical. The term angular distortion is used to indicate the deformation of the structure in a vertical plane.

5.The maximum safe bearing capacity of hard rocks with defects and lamination such as granite, diorite and trap is _____ kN/m². a) 1650

b) 3300

c) 250

d) 100

[View Answer](#)

Answer: b

Explanation: The term maximum safe bearing capacity of soil is used to indicate the maximum pressure that soil can bear here without any risk of shear failure only, irrespective of any settlement that may occur.

6. The maximum safe bearing capacity of very soft, wet, pasty or muddy clay is ____ kN/m². a) 150

b) 100 c) 50 d) 25

[View Answer](#)

Answer: c

Explanation: As the given soil is very soft, wet, muddy clay, the soil possesses very least minimum safe bearing capacity. This is because, this soil possesses very less intermolecular attraction between the molecules hence it has low bearing capacity.

7. In case of non-cohesive soils such as sand and gravel, ____% reduction in allowable bearing capacity of the soil should be applied, if the water table is situated below or near the bearing surface of the soil.

a) 20% b) 50% c) 80% d) 100% [View Answer](#)

Answer: b

Explanation: However, if water table is situated below the bearing surface of the soil at a depth equal to the width of the foundation trench, no such reduction should be applied. But in this case, 50% reduction is applied. For intermediate depths of the water table, the proportion and percentage reduction should be applied.

8. Identify the method given below for the determination of the bearing power of soil. a) Plate load test method

b) Electrical resistivity method c) Seismic refraction method d) Geophysical method View Answer

Answer: a

Explanation: Plate load test method is one of the methods of determination of bearing power of soil. Plate load test method is the

improved point method and it is widely used for determination of the bearing capacity. Other methods for determination are: method of loading, method of dropping a weight, analytical methods and Presumptive method.

9. In which method, a square steel plate is used to determine the bearing power of soil. a) Plate load test

b) Method of loading c) Analytical method d) Penetration test View Answer

Answer: b

Explanation: In method of loading, a square steel plate is used whose size depends on practical conditions and it generally varies from 3000 mm² to 7500 mm². The plate should have sufficient thickness to withstand the maximum bending stresses due to probable loading.

10. The bearing capacity of soil is calculated in _____ units. a) gm/cc

b) kg/m³ c) kN/m² d) N/m View Answer

Answer: c

Explanation: The bearing capacity of soil is calculated as the maximum load divided by the total area of steel plate i.e. Bearing Capacity = max. Load ÷ area of steel plate and the unit is N/m² or kN/m²

1. Foundations can be broadly classified under _____

a) Shallow foundation and Deep foundation b) Pile foundation

c) None of the mentioned d) All of the mentioned View Answer

Answer: a

Explanation: Foundations may be broadly classified under two heads: shallow foundations and deep foundations.

2. A foundation is said to be shallow if its depth is _____ than its width. a) Equal to and Less than

b) Greater than

c) None of the mentioned d) All of the mentioned View Answer

Answer: a

Explanation: According to Terzaghi, a foundation is shallow if its depth is equal to or less than its width. In the case of deep foundations, the depth is equal to or greater than the width.

3. Which of the following, is a type of shallow footing? a) Spread footing

b) Pile foundation c) Pier foundation

d)Well foundation View Answer

Answer: a

Explanation: The shallow foundations are of the following types: spread footing (or simple footing), strap footing, combined footing, and mat footing or raft footing.

4.Which of the below is the most commonly used shallow foundation? a) Strap footing

b) Spread footing

c) Combined footing d) Raft footing View Answer

Answer: b

Explanation: In spread footing, load is transmitted through an isolated column or wall to the subsoil; hence this is most common type of foundation.

5.The pressure intensity beneath the footing depends upon _____

a) Rigidity of the footing b) Soil type

c) Condition of soil

d) All of the mentioned View Answer

Answer: d

Explanation: Both from observations as well as the analytical studies from elasticity, it is known that the pressure distribution beneath footing is not uniform and it depends on the rigidity of footing, the soil type, and the condition of the soil.

6. Once the pressure distribution is known _____ in the reinforced concrete footing can be calculated. a) Bending moment and Shear force

b) Bearing pressure

c) None of the mentioned

d) All of the mentioned View Answer

Answer: a

Explanation: Knowing the pressure distribution is known, the bending moment and shear force can be calculated, and the thickness of the structural member of the member can be calculated using properties of reinforced concrete.

7. In conventional design, allowable bearing capacity should be taken smaller than which of the following value? a) Safe bearing capacity and Allowable bearing pressure

b) The pressure intensities beneath the footing c) None of the mentioned

d) All of the mentioned View Answer

Answer: a

Explanation: In conventional design, the allowable bearing capacity should be taken as the smaller of the following two values: i) the safe bearing capacity based on ultimate capacity, and ii) the allowable bearing pressure on tolerable settlement.

8. In cohesive soil, the pressure distribution beneath the footing is _____

a) Linear

b) Non linear c) Zero

d) None of the mentioned View Answer

Answer: a

Explanation: When a footing rests on cohesive soil, the edge stresses may be very large, but the pressure distribution may be considered to be linear.

9. When do strap footings are used in foundation? a) To transfer load of an isolated column

b) Distance between the columns are long c) Two column loads are unequal

d) All of the mentioned View Answer

Answer: b

Explanation: A strap footing may be used where the distance between the columns is so great that the combined trapezoidal footing becomes quite narrow, with a high bending moment.

10. When two column loads are unequal, which of the possible footing can be provided? a) Strap footing

b) Raft footing

c) Trapezoidal combined footing d) Mat footing

View Answer

Answer: c

Explanation: When the two columns load are unequal, with the outer column carrying a heavier load, and when there is space limitation beyond the outer column, a trapezoidal footing is provided

1. When compacted dry of optimum, the structure of clay is always ____

a) Flocculated

b) Dispersed

c) Disoriented

d) Honey comb-shaped

Correct Answer: a (As the structure of compacted clay is complicated and increasingly oriented, the structure of clay is always flocculated)

2. The flocculated structure of compacted dry soil is broken due to ____

a) High strains

b) Low water content

c) Low strains

d) Dry density

Answer: a (Due to higher strains, the flocculated structure of the compacted on the dry side is broken, giving rise to ultimate strength)

3. Strength of soil sample compacted wet of the optimum is influenced by ____

a) Compressibility

b) Permeability

c) Manner of compaction

d) Amount of compaction

Correct Answer: c (The manner of compaction influences the strength of soil sample compacted wet of optimum to a certain extent)

4. A cohesive soil yields a maximum dry density of 1.8 g/cc at an OMC of 16 % during a standard proctor test. What will be its degree of saturation? Take $G=2.65$

a) 100 %

b) 60.45 %

c) 43.27 %

d) 89.79 %

Correct Answer: d (Given $\rho_d = 1.8 \text{ g/cm}^3$; $w=0.16$; $G=2.65$, $e = G \rho_w / \rho_d - 1 = (2.65 \times 1 / 1.8) - 1 = 0.4722$, $S = w G / e = 0.16 \times 2.65 / 0.4722 = 0.8979 = 89.79 \%$, Degree of saturation, $S=89.79 \%$)

5. The shear strength of compacted clays depends upon _____

a) Dry density

b) Water content

c) Degree of saturation

d) Addition of admixtures

Correct Answer: a (The shear strength of compacted clays depends upon dry density, moulding water content, soil structure, method of compaction, drainage condition and type of soil)

6. The structure of composite soil after compacted will be in the form of _____

a) Coarse-grained skeleton structure and Cohesive matrix structure

b) Single grained structure

c) None of the mentioned

d) All of the mentioned

Correct Answer: a (The structure of composite soils, after being compacted depends upon the relative proportion of coarse particles their structure can either be coarse-grained skeleton structure or cohesive matrix structure)

7. What is the maximum dry density for a soil sample having sp. gr. of 2.7 and OMC=16 %?

a) 3.0 g/cm³

b) 1.88 g/cm³

c) 0.562 g/cm³

d) 1.00 g/cm³

Correct Answer: b (when $S=1$, γ_d is maximum. so, $\gamma_d \text{ max} = \frac{G \gamma_w}{1 + (w \cdot G/S)} = \frac{G \gamma_w}{1 + w \cdot G}$ (since $S=1$) Hence, $\rho_{d \text{ max}} = \frac{G \cdot \rho_w}{1 + w \cdot G} = \frac{2.7 \times 1}{1 + (0.16 \times 2.7)}$ Maximum dry density (MDD), $\gamma_d \text{ max} = 1.885 \text{ g/cm}^3$.)